



Python Class Note: Operators



Objective:

By the end of this class, you should be able to:

- Understand what operators are in Python
- Identify different categories of operators
- Apply them correctly in expressions and programs



1. What Are Operators?

Definition:

Operators are **symbols** or **keywords** used to perform operations on variables and values.

For example:

```
x = 10
y = 5
print(x + y)    # Output: 15
```

Here, **+** is an **operator** that adds `x` and `y`.



2. Types of Operators in Python

Python operators are grouped into **seven main categories**:

Category	Description	Examples	
Arithmetic Operators	Perform mathematical operations	<code>+</code> , <code>-</code> , <code>*</code> , <code>/</code> , <code>%</code> , <code>//</code> , <code>**</code>	
Comparison Operators	Compare two values	<code>==</code> , <code>!=</code> , <code>></code> , <code><</code> , <code>>=</code> , <code><=</code>	

Category	Description	Examples	
Assignment Operators	Assign values to variables	= , += , -= , *= , /= , //= , **= , %=	
Logical Operators	Combine conditional statements	and , or , not	
Bitwise Operators	Perform bit-level operations	& , ^ , ~ , << , >>	
Membership Operators	Test for membership in a sequence	in , not in	
Identity Operators	Compare memory locations of objects	is , is not	

+ 3. Arithmetic Operators

Operator	Description	Example	Output
+	Addition	10 + 5	15
-	Subtraction	10 - 5	5
*	Multiplication	10 * 5	50
/	Division (float)	10 / 3	3.333...
//	Floor Division	10 // 3	3
%	Modulus (Remainder)	10 % 3	1
**	Exponentiation	2 ** 3	8

Example:

```
a, b = 10, 3
print(a + b, a - b, a * b, a / b, a // b, a % b, a ** b)
```

Class Exercise:

Write a program to calculate the area and perimeter of a rectangle using arithmetic operators.



4. Comparison Operators

These operators compare **two values** and return a **Boolean** (True or False).

Operator	Description	Example	Output
==	Equal to	5 == 5	True
!=	Not equal to	5 != 3	True
>	Greater than	10 > 2	True
<	Less than	10 < 2	False
>=	Greater than or equal to	5 >= 5	True
<=	Less than or equal to	5 <= 10	True



Example:

```
x, y = 7, 10
print(x > y)
print(x == y)
print(x != y)
```



Class Exercise:

Write a Python script to check if a person is eligible to vote (age $\geq$ 18).



5. Assignment Operators

Used to **assign values** to variables, often in combination with arithmetic operations.

Operator	Example	Equivalent To
=	x = 5	x = 5
+=	x += 3	x = x + 3
-=	x -= 2	x = x - 2

Operator	Example	Equivalent To
<code>*=</code>	<code>x *= 2</code>	<code>x = x * 2</code>
<code>/=</code>	<code>x /= 2</code>	<code>x = x / 2</code>
<code>%=</code>	<code>x %= 3</code>	<code>x = x % 3</code>
<code>**=</code>	<code>x **= 2</code>	<code>x = x ** 2</code>
<code>//=</code>	<code>x //= 3</code>	<code>x = x // 3</code>

Example:

```
x = 5
x += 2
print(x)    # Output: 7
```

Class Exercise:

Write a program that takes an integer and doubles it using an assignment operator.

6. Logical Operators

Used to combine multiple **conditions**.

Operator	Description	Example	Result
and	Returns True if both are true	<code>(5 > 2) and (10 > 5)</code>	True
or	Returns True if at least one is true	<code>(5 > 2) or (10 < 5)</code>	True
not	Reverses the result	<code>not(5 > 2)</code>	False

Example:

```
age = 20
has_id = True
print(age >= 18 and has_id)
```



Class Exercise:

Write a program to check if a number is **between 10 and 50** (inclusive).



7. Bitwise Operators

Operate on numbers at the **binary level** (0s and 1s).

Operator	Description	Example	Result (Decimal)		
&	AND	5 & 3	1		
	OR	5 3	7		
^	XOR	5 ^ 3	6		
~	NOT	~5	-6		
<<	Left shift	5 << 1	10		
>>	Right shift	5 >> 1	2		



Example:

```
x = 5
y = 3
print(x & y, x | y, x ^ y)
```



Class Exercise:

Print the binary representation of 12 and 6, then compute `12 & 6`, `12 | 6`, and `12 ^ 6`.



8. Membership Operators

Used to check whether a **value exists** in a sequence (like list, tuple, string, etc.).

Operator	Description	Example	Result
in	Returns True if value is present	'a' in 'apple'	True

Operator	Description	Example	Result
not in	Returns True if value is absent	'b' not in 'apple'	True

Example:

```
fruits = ['apple', 'banana', 'cherry']  
print('apple' in fruits)  
print('grape' not in fruits)
```

Class Exercise:

Write a program that checks if a user's input exists in a predefined list of usernames.

9. Identity Operators

Used to check whether **two variables refer to the same object** in memory.

Operator	Description	Example	Result
is	True if both refer to the same object	a is b	Depends
is not	True if they refer to different objects	a is not b	Depends

Example:

```
a = [1, 2, 3]  
b = a  
c = [1, 2, 3]  
  
print(a is b)    # True  
print(a is c)    # False  
print(a == c)    # True (values are equal)
```

Class Exercise:

Write a program that compares two lists using both `==` and `is` to show the difference.



10. Summary Table

Category	Example	Description	
Arithmetic	+ - * /	Mathematical calculations	
Comparison	== != > <	Compare values	
Assignment	= += -=	Assign values	
Logical	and or not	Combine conditions	
Bitwise	`&	^ << >>`	Binary operations
Membership	in, not in	Test for presence in a sequence	
Identity	is, is not	Compare memory references	



Final Class Activity

Write a program that:

- 1. Takes two numbers as input
- 2. Compares them
- 3. Performs all arithmetic operations
- 4. Checks if the numbers are in a list [5, 10, 15, 20]
- 5. Prints all the results